

Community Library Membership & Book Lending System

Overview

The Community Library Membership & Book Lending System is a .NET Core Console Application that uses Entity Framework Core with PostgreSQL to manage library operations. The application provides features for managing members, books, book copies, borrowing, returning, fine management, reporting, and business rule validation.

Objectives

The application should provide the following capabilities:

- Member Management
 - Book Management
 - Book Copy Management
 - Borrowing Books
 - Returning Books
 - Fine Management
 - Reports
 - PostgreSQL Function Integration
 - Transaction Management
-

Member Management

The system should allow librarians to manage library members.

Features

- Add a new member
- View all members
- Search members using phone number
- Search members using email address
- Update membership status
- Deactivate a member

Membership Types

Basic

- Maximum Active Borrowings: 2 books
- Maximum Borrow Duration: 7 days

Student

- Maximum Active Borrowings: 3 books
- Maximum Borrow Duration: 10 days

Premium

- Maximum Active Borrowings: 5 books
 - Maximum Borrow Duration: 15 days
-

Book Management

The application should manage books available in the library.

Features

- Add new books
 - Add multiple copies of a book
 - View available books
 - Search books by title
 - Search books by author
 - Search books by category
 - Mark a book copy as damaged
 - Mark a book copy as unavailable
-

Book Categories

Books belong to categories such as:

- Programming
- Database
- Networking

- Artificial Intelligence
- Cloud Computing
- Operating Systems
- Data Science

Each book belongs to one category.

Book Copies

A single book can have multiple physical copies.

Each copy has its own availability status.

Possible statuses include:

- Available
- Borrowed
- Damaged
- Unavailable

Only available copies can be borrowed.

Borrowing Process

When a member requests a book, the application validates several business rules before approving the borrowing request.

Borrowing Rules

The member must have an active membership.

The member must not exceed the borrowing limit defined by the membership type.

The selected book copy must be available.

The member must not have unpaid fines greater than ₹500.

The member must not already have an active borrowing for the same book.

If all validations succeed:

1. Create a borrowing record.
2. Mark the selected book copy as borrowed.
3. Save the borrowing date.
4. Calculate the expected return date based on membership type.
5. Commit the transaction.

If any validation fails, the transaction must be rolled back.

Borrowing Limits

Membership Type	Maximum Books	Borrow Days
-----------------	---------------	-------------

Basic	2	7
Student	3	10
Premium	5	15

Return Process

When a member returns a book:

- Save the return date.
- Calculate late days.
- Calculate fine if applicable.
- Update borrowing status to Returned.
- Mark the book copy as Available.

Fine Calculation

Late return fine is calculated using the following rule:

$\text{Fine} = ₹10 \times \text{Number of Delayed Days}$

Examples:

If delayed by 3 days:

$\text{Fine} = ₹30$

If delayed by 8 days:

Fine = ₹80

If returned on time:

Fine = ₹0

Fine Management

The system allows librarians to manage fines.

Features

- View pending fines
 - Pay fines
 - View fine payment history
 - Prevent borrowing if unpaid fine exceeds ₹500
-

Borrowing Status

Borrowing records may have the following statuses:

- Active
 - Returned
-

Book Copy Status

Each book copy can be in one of the following states:

- Available
 - Borrowed
 - Damaged
 - Unavailable
-

Database Entities

The application contains several entities.

Possible entities include:

- Member
- MembershipType
- Book
- BookCategory
- BookCopy
- Borrowing
- Fine
- FinePayment

Relationships should be designed using Entity Framework Core.

PostgreSQL Function

The application should include at least one PostgreSQL function.

Recommended function:

`calculate_member_fine(member_id)`

Purpose:

Returns the total unpaid fine amount for a member.

Additional suggested functions:

- `get_available_books_by_category(category_id)`
 - `get_member_borrowing_summary(member_id)`
-

Transaction Management

Borrowing a book must be executed as a database transaction.

Transaction Steps:

1. Validate member.
2. Validate unpaid fines.
3. Check borrowing limit.
4. Verify book availability.
5. Create borrowing record.
6. Update book copy status.
7. Commit transaction.

If any step fails:

- Rollback the transaction.
-

Reports

The application should generate useful reports.

Available reports include:

- Books currently borrowed
 - Overdue books
 - Members with pending fines
 - Most borrowed books
 - Available books by category
 - Member borrowing history
-

Validation Rules

The application should validate:

- Invalid member
 - Invalid book
 - Invalid book copy
 - Inactive membership
 - Borrowing limit exceeded
 - Fine limit exceeded
 - Duplicate active borrowing
 - No available book copy
-

Console Menu

The application should provide the following menu options:

1. Member Management
 2. Book Management
 3. Borrow Book
 4. Return Book
 5. Fine Management
 6. Reports
 7. Exit
-

Testing Scenarios

The application should be tested using the following scenarios:

- Borrow books using Basic membership.
 - Borrow books using Student membership.
 - Borrow books using Premium membership.
 - Attempt borrowing after reaching the borrowing limit.
 - Attempt borrowing with unpaid fines above ₹500.
 - Return books before the due date.
 - Return books after the due date.
 - Verify fine calculation.
 - Verify transaction rollback.
 - Verify PostgreSQL function execution.
 - Verify report generation.
-

Summary

The Community Library Membership & Book Lending System demonstrates the implementation of Entity Framework Core with PostgreSQL while enforcing real-world business rules. It includes CRUD operations, transaction management, borrowing workflows, return processing, fine calculation, reporting, validation, and PostgreSQL function integration.